

Renal and Urinary Health

Filtration, tubular transport, fluid, electrolytes, acid-base, endocrine
kidney function, urinary tract, dialysis, and renal devices

Volume 03F of the Health Atlas Research Suite

Purpose

Connect nephron and urinary physiology to longitudinal risk, laboratory inference, imaging, dialysis and device engineering while keeping whole-person function, treatment burden and cardiorenal-metabolic coupling explicit.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

Evidence and guidance are current through 13 August 2026. This volume does not provide individual fluid, electrolyte, dialysis or medication-dosing instructions, nor instructions for constructing a dialysis machine or invasive urine catheter system. Severe electrolyte or acid-base abnormalities, infected obstruction, rapidly progressive kidney injury and dialysis-device faults require professional response.

How to use this volume

Kidney health is the capacity to filter plasma, reclaim and secrete solutes, regulate water, electrolytes and acid-base, support pressure and endocrine functions, handle drugs and toxins, and store and eliminate urine safely. Creatinine, eGFR and urine output are useful proxies but are not the organ itself. A complete profile separates filtration from injury, albumin leak, tubular function, volume, electrolytes, endocrine consequences, obstruction/infection, urinary function, reserve and trajectory. It also accounts for cardiovascular and metabolic causes and consequences, medication exposure, muscle/body composition, life stage, access and the lived burden of dialysis or incontinence. Renal devices are high-risk human-machine systems where calibration, fluid pathways, independent alarms, cybersecurity and fallback care are part of physiology and safety.

Questions this volume is designed to answer

1. What do filtration, injury, albuminuria, urine output and volume each represent?
2. How do kidney, heart, endocrine, GI and respiratory systems jointly regulate fluid and acid-base?
3. How should AKI and CKD be classified without overinterpreting estimates?
4. What can urine, imaging, biopsy and urodynamic tests establish?
5. How should home renal and dialysis devices be calibrated, fault-tested and governed?
6. Which interventions prevent progression and reduce cardiorenal outcomes?
7. How should function, treatment burden, conservative care and equity be represented?

Scope

- Glomerular filtration, tubular transport, water, sodium, potassium, acid-base and endocrine renal functions.
- Urinary tract transport, storage, voiding, defense, drug/toxin handling, repair and reserve.
- AKI, CKD, diabetic, glomerular, interstitial, genetic, stone, infectious, obstructive, electrolyte, urinary and cancer phenotypes.
- Creatinine/cystatin/measured GFR, urine testing, imaging, biopsy, urodynamics and novel biomarkers.
- Dialysis, urine-output, bladder and home-monitoring device engineering and validation.
- Prevention, pharmacology, urologic procedures, dialysis, transplant, rehabilitation and conservative care.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

Contents

Document status and intended use	i
How to use this volume	ii
1 Executive synthesis	1
1.1 What a complete profile requires	1
2 Historical development and changing models	3
2.0.1 Urine volume, color, taste/specific gravity, edema, stones and urinary obstruction were core observations.	4
2.0.2 Microscopy, chemical urinalysis, Bright disease and renal pathology linked edema/proteinuria to kidney lesions.	4
2.0.3 Clearance physiology, acid-base chemistry, blood pressure science, antibiotics and early dialysis emerged.	4
2.0.4 Chronic hemodialysis, peritoneal dialysis, transplantation, immunology, renal biopsy and urodynamics matured.	4
2.0.5 ACE inhibition, standardized creatinine, eGFR equations, albuminuria and modern ultrasound/CT transformed CKD detection.	5
2.0.6 Cystatin C, automated urine microscopy, kidney genetics, home dialysis technology and cardiorenal therapies expanded.	5
2.0.7 SGLT2 and related cardiorenal treatment, removal of race coefficients, remote dialysis monitoring and AI imaging/models accelerated.	5
2.0.8 Kidney health includes filtration, volume, electrolytes, acid-base, endocrine function, toxin/drug handling, urinary storage/voiding and reserve under stress.	5
3 Normal foundations: structure, physiology, and regulation	7
3.0.1 Glomerular filtration	7
3.0.2 Tubular transport	7
3.0.3 Water and osmolality	8
3.0.4 Sodium, volume and pressure regulation	8
3.0.5 Potassium homeostasis	8
3.0.6 Acid-base regulation	9
3.0.7 Endocrine kidney functions	9
3.0.8 Urinary tract transport and defense	9
3.0.9 Drug and toxin handling	10
3.0.10 Nephron reserve, repair and fibrosis	10
3.1 Useful quantitative relationships	11
4 Determinants, heterogeneity, and causal architecture	12

4.0.1	Genetic architecture and family history	13
4.0.2	Development and life course	14
4.0.3	Nutrition and metabolism	14
4.0.4	Physical activity, conditioning, and sedentary behavior	14
4.0.5	Sleep, circadian timing, and recovery	15
4.0.6	Stress, emotion, and behavior	15
4.0.7	Social, economic, cultural, and care conditions	15
4.0.8	Environmental and occupational exposures	16
4.0.9	Health care, medication, and iatrogenic effects	16
5	Major disorders, failure modes, and clinically important phenotypes	17
5.0.1	Acute kidney injury	17
5.0.2	Chronic kidney disease	17
5.0.3	Diabetic kidney disease	18
5.0.4	Glomerulonephritis and nephritic syndromes	18
5.0.5	Nephrotic syndrome and proteinuric disease	19
5.0.6	Tubulointerstitial and drug-induced kidney disease	19
5.0.7	Polycystic and inherited kidney disease	20
5.0.8	Kidney stones	20
5.0.9	Urinary tract infection and pyelonephritis	21
5.0.10	Obstructive uropathy	21
5.0.11	Electrolyte disorders	22
5.0.12	Acid-base disorders	22
5.0.13	Kidney failure and uremic syndrome	23
5.0.14	Dialysis complications	23
5.0.15	Lower urinary tract dysfunction and incontinence	24
5.0.16	Renal and urinary cancers	24
6	Building a kidney and fluid-monitoring platform	26
6.1	Low-risk longitudinal stack	26
6.2	Signals that must not be conflated	26
7	Dialysis device safety architecture	27
7.1	Critical hazards	27
8	Acid-base as a coupled-system problem	28
8.1	Minimum reasoning sequence	28
9	Assessment, clinical reasoning, and longitudinal profiling	29
9.0.1	1. Question and urgency	30
9.0.2	2. History and lived phenotype	30

9.0.3	3. Examination and basic measurements	30
9.0.4	4. Targeted laboratory, imaging, and physiology	30
9.0.5	5. Function and challenge testing	31
9.0.6	6. Longitudinal integration	31
10	Measurement and device landscape	32
10.0.1	History, pressure, volume and fluid balance	35
10.0.2	Serum creatinine and eGFR	35
10.0.3	Cystatin C and combined eGFR	35
10.0.4	Measured GFR and clearance tests	36
10.0.5	Urinalysis dipstick	36
10.0.6	Urine microscopy	37
10.0.7	Albumin/protein quantification	37
10.0.8	Urine electrolytes, osmolality and metabolic collections	37
10.0.9	Renal and bladder ultrasound	38
10.0.10	CT and CT urography	38
10.0.11	MRI and MR urography	38
10.0.12	Nuclear renography	39
10.0.13	Kidney biopsy	39
10.0.14	Bladder scan, uroflow and urodynamics	40
10.0.15	Continuous urine-output sensors	40
10.0.16	Dialysis machine and water-system sensing	40
10.0.17	Novel injury and molecular biomarkers	41
11	Engineering architectures, signal quality, and validation	42
11.0.1	Intended use and measurand	43
11.0.2	Sensor and transduction	43
11.0.3	Timing and synchronization	43
11.0.4	Analog, acquisition, and digital integrity	44
11.0.5	Calibration and traceability	44
11.0.6	Artifact and quality channels	44
11.0.7	Algorithms and uncertainty	45
11.0.8	Safety, usability, cybersecurity, and privacy	45
12	Prevention, treatment, rehabilitation, and system intervention	46
12.0.1	Blood pressure, diabetes and vascular prevention	48
12.0.2	RAAS blockade and antiproteinuric therapy	48
12.0.3	SGLT2 and cardiorenal-metabolic therapy	48
12.0.4	Volume, electrolyte and acid-base management	49
12.0.5	Immunologic and cause-specific kidney therapy	49

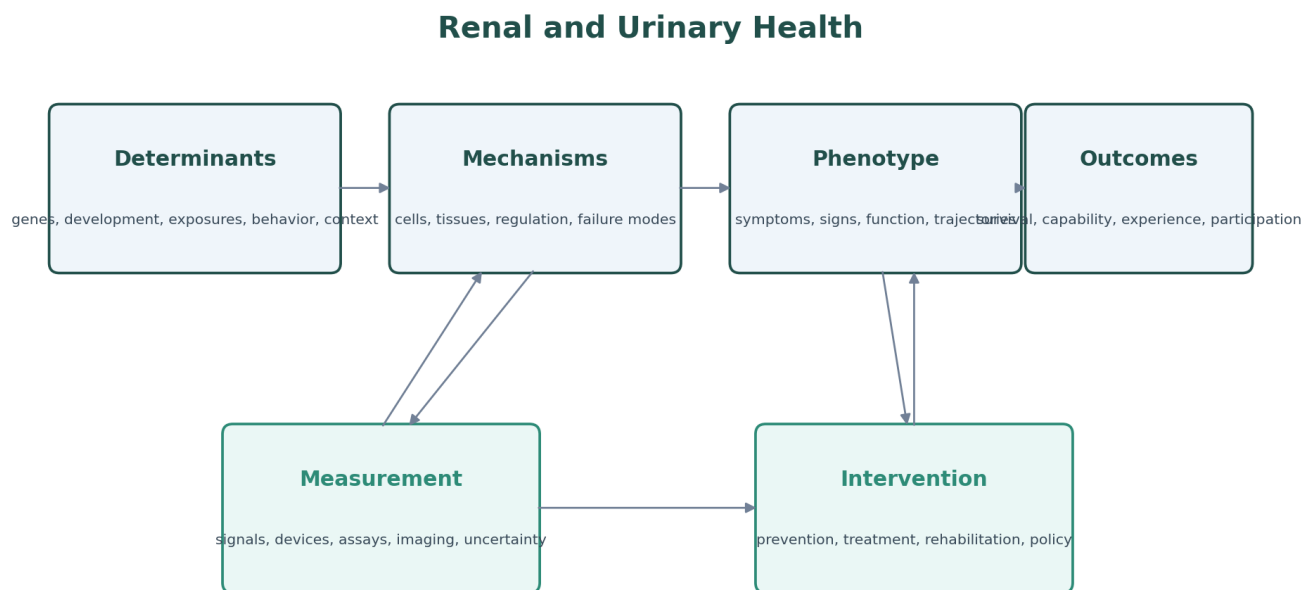
12.0.6 Medication stewardship and nephrotoxin prevention	49
12.0.7 Stone prevention and urologic procedure	50
12.0.8 Urinary infection prevention and treatment	50
12.0.9 Pelvic-floor and bladder rehabilitation	50
12.0.10 Hemodialysis and peritoneal dialysis	51
12.0.11 Kidney transplantation	51
12.0.12 Conservative kidney management and palliative care	51
13 What is known, not known, and frequently overstated	52
13.0.1 Can AKI biomarkers improve outcomes?	53
13.0.2 What is the best measure of volume status?	54
13.0.3 When should dialysis start and how intensive should it be?	54
13.0.4 Can home urine and electrolyte sensors be trustworthy?	54
13.0.5 How much risk is genetic versus environmental?	55
13.0.6 Can one composite score summarize health?	55
13.0.7 When is earlier detection beneficial?	55
13.0.8 Do digital biomarkers generalize?	55
13.0.9 What counts as meaningful change?	56
14 Research and development roadmap	57
14.0.1 0. Nonsteady filtration model	58
14.0.2 0. Fluid-status utility trial	58
14.0.3 0. Home dialysis resilience study	59
14.0.4 1. Reference map	59
14.0.5 2. Causal determinants	59
14.0.6 3. Measurement validity	59
14.0.7 4. Clinical utility	60
14.0.8 5. Longitudinal multimorbidity	60
14.0.9 6. Implementation and governance	60
15 Cross-volume integration	61
A Minimum data schema for a longitudinal domain profile	62
B Glossary	63
Selected references and current standards	64

Executive synthesis

Kidney health is the capacity to filter plasma, reclaim and secrete solutes, regulate water, electrolytes and acid-base, support pressure and endocrine functions, handle drugs and toxins, and store and eliminate urine safely. Creatinine, eGFR and urine output are useful proxies but are not the organ itself. A complete profile separates filtration from injury, albumin leak, tubular function, volume, electrolytes, endocrine consequences, obstruction/infection, urinary function, reserve and trajectory. It also accounts for cardiovascular and metabolic causes and consequences, medication exposure, muscle/body composition, life stage, access and the lived burden of dialysis or incontinence. Renal devices are high-risk human-machine systems where calibration, fluid pathways, independent alarms, cybersecurity and fallback care are part of physiology and safety.

Central organizing principle

Connect nephron and urinary physiology to longitudinal risk, laboratory inference, imaging, dialysis and device engineering while keeping whole-person function, treatment burden and cardiorenal-metabolic coupling explicit.



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.

- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Severe hyperkalemia, sodium disturbance, acidosis, pulmonary edema, anuria/obstruction with infection, rapidly worsening AKI or dialysis alarms can be life-threatening. Human dialysis and catheter systems require regulated design, independent safety controls, sterile fluid paths, water quality, electrical isolation and trained clinical oversight.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to 1800s	Urine volume, color, taste/specific gravity, edema, stones and urinary obstruction were core observations.	Established urine as a window into systemic disease and kidney function.	Filtration, nephron transport, infection and endocrine regulation were unknown.
1800s	Microscopy, chemical urinalysis, Bright disease and renal pathology linked edema/proteinuria to kidney lesions.	Created clinicopathologic nephrology and separated glomerular patterns.	Different diseases were grouped by gross syndromes and irreversible end-stage findings.
1900-1950	Clearance physiology, acid-base chemistry, blood pressure science, antibiotics and early dialysis emerged.	Quantified filtration, secretion/reabsorption and made acute organ support possible.	Clearance markers and dialysis were imperfect; chronic access and safety were limited.
1950-1980	Chronic hemodialysis, peritoneal dialysis, transplantation, immunology, renal biopsy and urodynamics matured.	Converted kidney failure from uniformly fatal to chronically treatable and clarified glomerular disease.	Treatment burden, infection, rejection and inequitable access became defining outcomes.
1980-2005	ACE inhibition, standardized creatinine, eGFR equations, albuminuria and modern ultrasound/CT transformed CKD detection.	Enabled risk staging and prevention of progression/vascular events.	Creatinine and race-based equations produced bias; CKD labels could overstate disease in low-risk aging.
2005-2020	Cystatin C, automated urine microscopy, kidney genetics, home dialysis technology and cardiorenal therapies expanded.	Improved confirmation, rare-disease diagnosis and integrated risk management.	Biomarkers often detected injury without proven utility; home therapies remained underused.
2020-2026	SGLT2 and related cardiorenal treatment, removal of race coefficients, remote dialysis monitoring and AI imaging/models accelerated.	Reduced progression and cardiovascular events in broader CKD groups and improved equity of estimation frameworks.	Implementation, acute safety, cost and calibration across populations remain.

Continued on next page

Period	Development	What it made possible	Residual limitation
Current systems view	Kidney health includes filtration, volume, electrolytes, acid-base, endocrine function, toxin/drug handling, urinary storage/voiding and reserve under stress.	Connects laboratory numbers to blood pressure, bone, anemia, cognition, pregnancy, function and lived treatment burden.	Care still separates nephrology, urology, hypertension, critical care and pelvic rehabilitation.

2.0.1 Urine volume, color, taste/specific gravity, edema, stones and urinary obstruction were core observations.

Context. Renal medicine began with longitudinal specimen observation.

Scientific consequence. Established urine as a window into systemic disease and kidney function.

What remained unresolved. Filtration, nephron transport, infection and endocrine regulation were unknown.

Evidence anchors: [5,8]

2.0.2 Microscopy, chemical urinalysis, Bright disease and renal pathology linked edema/proteinuria to kidney lesions.

Context. Urinalysis became a low-cost physiologic biopsy.

Scientific consequence. Created clinicopathologic nephrology and separated glomerular patterns.

What remained unresolved. Different diseases were grouped by gross syndromes and irreversible end-stage findings.

Evidence anchors: [5,8]

2.0.3 Clearance physiology, acid-base chemistry, blood pressure science, antibiotics and early dialysis emerged.

Context. The nephron became a quantitative transport system.

Scientific consequence. Quantified filtration, secretion/reabsorption and made acute organ support possible.

What remained unresolved. Clearance markers and dialysis were imperfect; chronic access and safety were limited.

Evidence anchors: [3,4,7]

2.0.4 Chronic hemodialysis, peritoneal dialysis, transplantation, immunology, renal biopsy and urodynamics matured.

Context. Renal replacement created complex human-device systems.

Scientific consequence. Converted kidney failure from uniformly fatal to chronically treatable and clarified glomerular disease.

What remained unresolved. Treatment burden, infection, rejection and inequitable access became defining outcomes.

Evidence anchors: [6,15]

2.0.5 ACE inhibition, standardized creatinine, eGFR equations, albuminuria and modern ultrasound/CT transformed CKD detection.

Context. Population surveillance grew around estimated rather than measured filtration.

Scientific consequence. Enabled risk staging and prevention of progression/vascular events.

What remained unresolved. Creatinine and race-based equations produced bias; CKD labels could overstate disease in low-risk aging.

Evidence anchors: [6,38]

2.0.6 Cystatin C, automated urine microscopy, kidney genetics, home dialysis technology and cardiorenal therapies expanded.

Context. Kidney health moved toward multimarker and patient-centered care.

Scientific consequence. Improved confirmation, rare-disease diagnosis and integrated risk management.

What remained unresolved. Biomarkers often detected injury without proven utility; home therapies remained underused.

Evidence anchors: [6,18]

2.0.7 SGLT2 and related cardiorenal treatment, removal of race coefficients, remote dialysis monitoring and AI imaging/models accelerated.

Context. Cardiovascular-kidney-metabolic medicine became a shared domain.

Scientific consequence. Reduced progression and cardiovascular events in broader CKD groups and improved equity of estimation frameworks.

What remained unresolved. Implementation, acute safety, cost and calibration across populations remain.

Evidence anchors: [2,6,9]

2.0.8 Kidney health includes filtration, volume, electrolytes, acid-base, endocrine function, toxin/drug handling, urinary storage/voiding and reserve under stress.

Context. A stable eGFR does not by itself prove stable nephron health, and an abnormal estimate is not a complete diagnosis.

Scientific consequence. Connects laboratory numbers to blood pressure, bone, anemia, cognition, pregnancy, function and lived treatment burden.

What remained unresolved. Care still separates nephrology, urology, hypertension, critical care and pelvic rehabilitation.

Evidence anchors: ^[6,8,17]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Glomerular filtration

Core principle. Hydrostatic and oncotic forces across a size/charge-selective barrier generate ultrafiltrate while retaining cells and most proteins.

Control architecture. Afferent/efferent tone, autoregulation, tubuloglomerular feedback, systemic pressure and filtration surface regulate GFR.

Failure modes. Hypoperfusion, obstruction, inflammation, thrombosis, podocyte/barrier injury or nephron loss reduces filtration or causes protein/hematuria.

Observable consequences. Creatinine/cystatin estimates, clearance, albuminuria, urine microscopy, imaging and measured GFR selectively.

Cross-system coupling. Cardiovascular pressure/flow, hematology, immunity, pregnancy and medications strongly affect filtration.

Evidence anchors: ^[3,4,6]

3.0.2 Tubular transport

Core principle. Segment-specific channels and transporters reclaim water/nutrients/electrolytes and secrete acids, potassium, drugs and toxins.

Control architecture. Hormones, flow, gradients, ATP and feedback coordinate reabsorption and secretion.

Failure modes. Ischemia, toxins, genetic transporter defects, drugs or interstitial disease causes electrolyte, glucose, acid-base and concentrating abnormalities.

Observable consequences. Serum/urine electrolytes, fractional excretion with caveats, urine pH/osmolality, glycosuria and tubular proteins.

Cross-system coupling. Endocrine, metabolic, pharmacologic and genetic systems converge.

Evidence anchors: ^[3,4,13]

3.0.3 Water and osmolality

Core principle. The kidney varies urine concentration to balance water while protecting plasma osmolality and volume.

Control architecture. Thirst, vasopressin, medullary gradient, collecting-duct channels and access to water regulate.

Failure modes. Diabetes insipidus, excess antidiuresis, low solute, polydipsia, loop disruption or kidney failure causes sodium/water disorders.

Observable consequences. Serum/urine osmolality and sodium, urine volume, volume status and water/deprivation testing selectively.

Cross-system coupling. Neurologic, endocrine, cardiac and behavioral systems are integral.

Evidence anchors: [2,3,10]

3.0.4 Sodium, volume and pressure regulation

Core principle. Renal sodium balance determines extracellular volume over time and participates in pressure control.

Control architecture. RAAS, sympathetic tone, natriuretic peptides, pressure natriuresis, aldosterone and nephron transport adapt excretion.

Failure modes. Nephron loss, endocrine excess, heart/liver disease, drugs or high intake can produce edema, hypertension or depletion.

Observable consequences. Pressure, weight/edema, sodium balance proxies, renin/aldosterone selectively and organ consequences.

Cross-system coupling. Cardiovascular, liver, endocrine and environmental nutrition systems couple.

Evidence anchors: [2,3,6]

3.0.5 Potassium homeostasis

Core principle. Cellular distribution and distal renal secretion maintain a narrow extracellular potassium range required for excitability.

Control architecture. Insulin, catecholamines, acid-base, aldosterone, distal sodium/flow and intake regulate.

Failure modes. Kidney failure, medications, endocrine disease, tissue breakdown or shifts cause arrhythmia and weakness.

Observable consequences. Serum potassium with hemolysis review, ECG in significant disturbance, urine indices selectively and causal context.

Cross-system coupling. Cardiac, neurologic, endocrine, muscle and drug systems determine risk.

Evidence anchors: [2,3,13]

3.0.6 Acid-base regulation

Core principle. Kidneys reclaim filtered bicarbonate, excrete titratable acid/ammonium and generate new bicarbonate over hours to days.

Control architecture. Respiratory carbon dioxide, tubular transport, ammoniagenesis, aldosterone and potassium balance interact.

Failure modes. Kidney failure, tubular defects, diarrhea, vomiting, toxins or respiratory disease produces mixed disturbances.

Observable consequences. Blood gas and electrolytes, anion gap, urine pH/electrolytes and clinical context.

Cross-system coupling. Respiratory and GI systems form the coupled acid-base system.

Evidence anchors: [3,4]

3.0.7 Endocrine kidney functions

Core principle. Kidneys produce renin and erythropoietin, activate vitamin D and clear/metabolize many hormones.

Control architecture. Oxygen sensing, perfusion, mineral feedback and nephron mass regulate.

Failure modes. CKD causes anemia, mineral-bone disorder, endocrine/drug accumulation and cardiovascular remodeling.

Observable consequences. Blood counts/iron, PTH/calcium/phosphate/vitamin D, pressure and medication pharmacokinetics.

Cross-system coupling. Hematologic, bone, endocrine and cardiovascular systems depend on renal function.

Evidence anchors: [6,11,12]

3.0.8 Urinary tract transport and defense

Core principle. Ureters, bladder, sphincters and pelvic floor store and expel urine while limiting reflux and infection.

Control architecture. Peristalsis, autonomic/somatic circuits, urothelium, flow, anatomy and behavior coordinate.

Failure modes. Obstruction, stones, neurogenic dysfunction, pelvic-floor disorder or infection causes retention, incontinence, reflux and kidney injury.

Observable consequences. Voiding history/diary, residual volume, urinalysis, imaging, flow and urodynamics selectively.

Cross-system coupling. Neurologic, reproductive, musculoskeletal, aging and social context determine function.

Evidence anchors: [8,16,17]

3.0.9 Drug and toxin handling

Core principle. Filtration, secretion, reabsorption and metabolism determine renal clearance and exposure to nephrotoxins.

Control architecture. Protein binding, transporter competition, urine pH, flow, GFR and dialysis regulate disposition.

Failure modes. Accumulation, crystal precipitation, hemodynamic effects, direct tubular injury or interactions causes toxicity.

Observable consequences. Medication/supplement exposure, creatinine/cystatin, urine, drug levels selectively and timing.

Cross-system coupling. Pharmacology, liver metabolism, hydration and genetics intersect.

Evidence anchors: [7,13,14]

3.0.10 Nephron reserve, repair and fibrosis

Core principle. Remaining nephrons adapt to loss, while injured tubules, glomeruli, vessels and interstitium may recover or scar.

Control architecture. Hemodynamic adaptation, epithelial repair, immune resolution, matrix turnover and risk-factor control shape trajectory.

Failure modes. Maladaptive hyperfiltration, recurrent AKI, ongoing inflammation or obstruction accelerates fibrosis and decline.

Observable consequences. eGFR slope, albuminuria, injury markers selectively, imaging, biopsy and reserve research tests.

Cross-system coupling. Aging, diabetes, pressure, immunity, environment and treatment timing drive progression.

Evidence anchors: [5,6,27]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Clearance	$C_x = \frac{U_x \dot{V}}{P_x}$	Virtual plasma volume cleared of substance x per unit time.	Requires accurate timed urine and steady concentration; secretion/reabsorption means clearance may not equal GFR.
Filtered load	Filtered load = $GFR \times P_x$	Amount of freely filtered solute entering tubules per time.	Protein binding, charge/size and non-steady GFR alter the approximation.
Excretion rate	Excretion = $U_x \dot{V}$	Urinary concentration times flow gives amount excreted per time.	Spot concentration without flow cannot establish excretion.
Net tubular handling	Excretion = Filtration + Secretion – Reabsorption	Mass-balance decomposition of nephron handling.	Multiple segments and metabolism complicate direct inference from urine and plasma.
Anion gap	$AG = Na^+ - (Cl^- + HCO_3^-)$	Screening proxy for unmeasured anions in acid-base assessment.	Albumin, assay, potassium convention and mixed disorders alter interpretation.
Cerebral/cardiac medication dosing exposure	Dose rate $\approx CL \times C_{target}$	Maintenance dosing relates desired concentration to total clearance.	Pharmacodynamics, active metabolites, dialysis, nonlinearity and individual targets require drug-specific guidance.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within renal and urinary health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within renal and urinary health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[18]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[38]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[34,39]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the renal syndrome, urinary dysfunction, homeostatic failure, or treatment state, but cross-system processes and functional effects are retained.

5.0.1 Acute kidney injury

Definition and phenotype. Abrupt decline in kidney filtration and homeostatic function, identified by creatinine and/or urine-output change and clinical context.

Pathophysiology. Hypoperfusion/hemodynamic change, intrinsic glomerular/tubular/interstitial/vascular injury or obstruction reduces filtration and disrupts transport.

Determinants and modifiers. Sepsis, surgery, shock, nephrotoxins, contrast in context, obstruction, heart/liver failure and baseline CKD.

Assessment. Baseline/trajectory of creatinine and urine output, volume/hemodynamics, medication/exposure, urinalysis/microscopy, ultrasound and targeted serology/biopsy when indicated.

Intervention logic. Treat cause, restore appropriate perfusion, stop/adjust harmful drugs, manage electrolytes/volume/nutrition and initiate kidney replacement for refractory life-threatening complications.

Known limits and open questions. Creatinine is delayed and affected by muscle/volume; biomarkers do not yet universally improve outcomes; fluid and timing strategies are context-specific.

Suite links. 04 critical illness; 03A circulation; 08 sepsis; 14 pharmacology.

Evidence anchors: [7,34,35]

5.0.2 Chronic kidney disease

Definition and phenotype. Abnormal kidney structure or function present long enough to imply chronicity, staged by cause, GFR and albuminuria with prognosis and function.

Pathophysiology. Nephron loss, glomerular/tubular/vascular disease and maladaptive hyperfiltration/fibrosis reduce reserve and cause systemic complications.

Determinants and modifiers. Diabetes, hypertension, immune/genetic disease, recurrent AKI, toxins, infection, obstruction, prematurity and social/environmental determinants.

Assessment. Establish chronicity and cause; creatinine eGFR with cystatin confirmation when useful; albuminuria, urine, imaging and complications; risk equations in intended populations.

Intervention logic. Cause and pressure/diabetes treatment, RAAS/SGLT2 and other indicated therapy, avoidance/adjustment of toxins/drugs, nutrition, vaccination, complication and kidney-failure planning.

Known limits and open questions. eGFR categories do not equal individual disease burden; older adults, muscle extremes and acute change require caution; implementation gaps dominate outcomes.

Suite links. 03A cardiovascular; 03D diabetes; 03G anemia; 07 aging.

Evidence anchors: [2,6,9]

5.0.3 Diabetic kidney disease

Definition and phenotype. Albuminuria and/or reduced filtration attributable in context to diabetes, often with vascular and metabolic comorbidity.

Pathophysiology. Glomerular hyperfiltration, metabolic stress, endothelial/podocyte injury, inflammation and fibrosis interact.

Determinants and modifiers. Glucose exposure, pressure, atherogenic risk, smoking, genetics, obesity, AKI and treatment access.

Assessment. Annual/appropriate eGFR and albuminuria, repeat confirmation, urine/clinical features for alternative disease and cardiovascular risk.

Intervention logic. Glucose and pressure control; RAAS blockade when indicated, SGLT2 and other cardiorenal therapies by eligibility, risk reduction and avoidance of AKI/medication harm.

Known limits and open questions. Not all CKD in diabetes is diabetic; regression of albuminuria does not guarantee no risk; optimal combination sequencing evolves.

Suite links. 03D metabolic; 03A cardiovascular; 14 therapeutics.

Evidence anchors: [2,6,9]

5.0.4 Glomerulonephritis and nephritic syndromes

Definition and phenotype. Immune, infectious, genetic or vascular glomerular injury produces hematuria, proteinuria, hypertension and variable rapid loss of filtration.

Pathophysiology. Immune complexes, antibodies, complement, inflammation or structural defects damage capillary wall and mesangium.

Determinants and modifiers. Autoimmune disease, infection, malignancy, drugs and inherited variants; some remain idiopathic.

Assessment. Urine microscopy/protein, kidney function, complement/serology/infection tests, biopsy and genetics when results change treatment.

Intervention logic. Cause-specific immune or anti-infective therapy, supportive pressure/proteinuria management and complication prevention; urgent therapy for rapidly progressive disease.

Known limits and open questions. Serologic positivity is not always causal; biopsy sampling and classifications evolve; immunosuppression benefits and harms vary by subtype.

Suite links. 03H immunity; 08 infection; 14 immunotherapy; 03A pressure.

Evidence anchors: [6,22,23]

5.0.5 Nephrotic syndrome and proteinuric disease

Definition and phenotype. Heavy protein loss with hypoalbuminemia and edema, often accompanied by hyperlipidemia and thrombosis/infection risk.

Pathophysiology. Podocyte or glomerular barrier failure increases protein permeability, altering oncotic and hepatic/renal responses.

Determinants and modifiers. Primary podocytopathies, diabetes, amyloid, autoimmune/infectious disease, drugs, genetic disease and cancer.

Assessment. Quantify protein/albumin, kidney function, sediment, secondary causes; biopsy/genetics by phenotype; assess edema, thrombosis, infection and nutrition.

Intervention logic. Cause-specific treatment; sodium/diuretic and RAAS strategies, selected immune therapy, thrombosis risk management and vaccination.

Known limits and open questions. Edema mechanisms and prophylactic anticoagulation thresholds vary; protein reduction is an imperfect surrogate.

Suite links. 03G thrombosis; 03A volume; 03H immune; 04 cancer.

Evidence anchors: [6,24]

5.0.6 Tubulointerstitial and drug-induced kidney disease

Definition and phenotype. Injury centered on tubules/interstitium causes concentrating, electrolyte and filtration abnormalities.

Pathophysiology. Toxins, immune hypersensitivity, infection, obstruction or metabolic deposits injure epithelium and interstitium.

Determinants and modifiers. NSAIDs/antibiotics/PPIs and other drugs, toxins, autoimmune disease, infection, reflux and genetic disorders.

Assessment. Medication/exposure timeline, urinalysis, tubular markers/electrolytes, imaging and biopsy when diagnosis changes management.

Intervention logic. Remove cause, treat infection/obstruction and selected immune disease, correct electrolytes and monitor recovery/CKD.

Known limits and open questions. Classic hypersensitivity features are often absent; biomarkers lack specificity; rechallenge is usually unsafe.

Suite links. 14 pharmacology; 03H immunity; 08 infection.

Evidence anchors: [7,14,34]

5.0.7 Polycystic and inherited kidney disease

Definition and phenotype. Genetic cystic, basement-membrane, transport or developmental disorders produce structural and multisystem kidney disease.

Pathophysiology. Variants alter cilia, matrix, channels, development or metabolism, causing cysts, stones, protein/hematuria or progressive loss.

Determinants and modifiers. Genotype, modifiers, blood pressure, sex/life stage, exposures and family context shape penetrance.

Assessment. Family/pedigree, imaging, urine/labs, extrarenal phenotype and genetics with counseling when useful.

Intervention logic. Pressure and complication management, selected disease-modifying therapy, family screening/counseling, kidney-failure planning and organ-specific surveillance.

Known limits and open questions. Variants of uncertain significance and incidental findings can harm; genotype does not precisely predict course.

Suite links. 02 genomics; 03A aneurysm/pressure; 07 life course; 15 genetics ethics.

Evidence anchors: [6,18,19]

5.0.8 Kidney stones

Definition and phenotype. Crystals aggregate in urinary tract and cause pain, obstruction, infection or recurrent kidney injury.

Pathophysiology. Supersaturation, low urine volume, pH, promoters/inhibitors, infection and anatomy determine nucleation/growth.

Determinants and modifiers. Fluid/diet, heat/work exposure, GI malabsorption, obesity/metabolic disease, medications, genetics and infection.

Assessment. Pain/infection/obstruction assessment; urinalysis, kidney function and imaging; stone analysis and 24-hour urine in recurrent/high-risk disease.

Intervention logic. Urgent drainage for infected obstruction; analgesia and passage/procedure strategy; fluid/diet and targeted medication by stone phenotype.

Known limits and open questions. Hydration advice must consider heart/kidney states; consumer urine sensors and generic diets can mislead.

Suite links. 03E malabsorption; 09 heat/occupation; 08 infection; 03D metabolism.

Evidence anchors: [6,8]

5.0.9 Urinary tract infection and pyelonephritis

Definition and phenotype. Microbial infection of bladder or kidney ranges from local symptoms to bacteremia/sepsis.

Pathophysiology. Ascending infection and host response interact with urinary stasis, reflux, stones, catheters and anatomy.

Determinants and modifiers. Sex/anatomy, pregnancy, diabetes, catheter, obstruction, prior antibiotics/resistance and immune state.

Assessment. Symptom and severity assessment; urinalysis/culture according to context; imaging for obstruction/complication; avoid treating asymptomatic bacteriuria except defined groups.

Intervention logic. Appropriate antibiotics by site/severity/resistance, source control and catheter/obstruction management; prevention by removing risk.

Known limits and open questions. Urine cultures often reflect colonization; recurrent symptoms can have noninfectious causes; resistance constrains empiric treatment.

Suite links. 08 infection/AMR; 03N pregnancy; 12 catheter safety.

Evidence anchors: [8,20,21]

5.0.10 Obstructive uropathy

Definition and phenotype. Blockage of urine flow from kidney to urethra can cause pain, infection, retention and reversible or permanent kidney injury.

Pathophysiology. Back pressure, altered perfusion and inflammation reduce filtration; chronic obstruction damages tubules and interstitium.

Determinants and modifiers. Stones, prostate, tumor, stricture, neurogenic bladder, pelvic organ prolapse, pregnancy and congenital anatomy.

Assessment. Urgency based on infection, pain, renal function and bilateral/solitary kidney; bladder residual and ultrasound/CT; localize cause.

Intervention logic. Prompt drainage when threatened/infected, then definitive treatment of cause; monitor post-obstructive diuresis and recovery.

Known limits and open questions. Hydronephrosis does not always equal obstruction; chronicity and recoverability are imperfectly predicted.

Suite links. 03M reproductive; 03C neurogenic bladder; 04 cancer; 12 acute care.

Evidence anchors: [8,34]

5.0.11 Electrolyte disorders

Definition and phenotype. Abnormal sodium, potassium, magnesium, calcium or phosphate reflects intake, distribution, kidney handling, endocrine control and losses.

Pathophysiology. Changes in water balance, transport, hormones, cell shifts or organ failure alter extracellular concentrations and excitability.

Determinants and modifiers. Kidney/heart/liver/endocrine/GI disease, medications, nutrition, exercise/heat and acute illness.

Assessment. Confirm specimen and rate; symptoms/ECG where relevant; serum/urine and acid-base context; medication and volume assessment.

Intervention logic. Treat severity and cause while controlling correction rate; stop/adjust drugs, replace/remove solute and monitor closely.

Known limits and open questions. Concentration is not total-body store; formulas are approximations; overrapid correction can cause neurologic harm.

Suite links. 03D endocrine; 03A cardiac; 03C neurologic; 03E GI losses.

Evidence anchors: [3,6,34]

5.0.12 Acid-base disorders

Definition and phenotype. Metabolic acidosis/alkalosis and mixed disturbances reflect kidney, respiratory, GI, metabolic and toxin processes.

Pathophysiology. Bicarbonate, carbon dioxide and unmeasured acids change through production, excretion and compensation.

Determinants and modifiers. Kidney failure/tubular disease, diarrhea/vomiting, keto/lactic acidosis, toxins, ventilation and medications.

Assessment. Blood gas, electrolytes, anion gap adjusted for context, compensation checks and targeted urine/toxin/lactate/ketone evaluation.

Intervention logic. Treat cause and severe physiology; bicarbonate/ventilatory/dialysis strategies only in defined contexts; monitor correction.

Known limits and open questions. Venous/arterial differences and formulas can mislead; mixed disorders are common.

Suite links. 03B respiratory; 03D metabolism; 03E GI; 04 critical illness.

Evidence anchors: [3,7,35]

5.0.13 Kidney failure and uremic syndrome

Definition and phenotype. Advanced loss of kidney function causes fluid, electrolyte, acid-base, endocrine and toxin-related symptoms requiring conservative care, dialysis or transplant.

Pathophysiology. Insufficient filtration/transport and endocrine function produces multisystem accumulation and dysregulation.

Determinants and modifiers. Progressive CKD, recurrent AKI, limited access and comorbid disease.

Assessment. Symptoms, nutrition/function, volume/electrolytes, eGFR trajectory and individualized planning; dialysis initiation is not based on one number alone.

Intervention logic. Comprehensive conservative care or hemodialysis/peritoneal dialysis/transplant according to goals, eligibility and access; symptom and palliative support.

Known limits and open questions. “Uremic toxins” are heterogeneous; optimal modality/timing/frequency and patient selection depend on outcomes and preferences.

Suite links. 12 delivery systems; 10 function; 15 equity; 14 devices.

Evidence anchors: [6,36,37]

5.0.14 Dialysis complications

Definition and phenotype. Extracorporeal or peritoneal therapy can cause hemodynamic, infectious, access, membrane, electrolyte and treatment-burden complications.

Pathophysiology. Rapid solute/fluid shifts, blood-material interaction, vascular access and peritoneal exposure create nonphysiologic stress.

Determinants and modifiers. Frailty, cardiac disease, residual function, access type, prescription, water quality, staffing and home support.

Assessment. Continuous treatment data plus symptoms, pressure, volume, labs, access/peritoneum and function; investigate alarms and machine-water systems.

Intervention logic. Prescription and ultrafiltration optimization, access/infection prevention, home support, modality change and goals-based care.

Known limits and open questions. Small-solute adequacy does not equal well-being or middle-molecule/volume control; device telemetry can miss lived burden.

Suite links. 03A hemodynamics; 08 infection; 12 system safety; 10 function.

Evidence anchors: [6,30,34]

5.0.15 Lower urinary tract dysfunction and incontinence

Definition and phenotype. Storage or voiding symptoms arise from bladder, outlet, pelvic-floor, neurologic, infection, medication or behavioral causes.

Pathophysiology. Detrusor activity/compliance, sphincter coordination, sensation and anatomy become mismatched.

Determinants and modifiers. Age, childbirth, prostate disease, neurologic disease, diabetes, mobility/cognition, medications and environment/toilet access.

Assessment. Bladder diary, urinalysis, residual volume, pelvic/neurologic exam; urodynamics or imaging when results change invasive treatment.

Intervention logic. Behavior/fluid and medication review, pelvic-floor therapy, bladder training, drugs or procedures by subtype, continence products and environmental accommodation.

Known limits and open questions. Symptoms poorly predict urodynamic mechanism; overtreatment and anticholinergic burden can harm cognition.

Suite links. 03C neurologic; 03I pelvic floor; 03M reproductive; 10 disability.

Evidence anchors: [8,16,17]

5.0.16 Renal and urinary cancers

Definition and phenotype. Kidney, renal pelvis, bladder or urinary tract malignancies vary by cell type, exposure and stage.

Pathophysiology. Somatic evolution, tobacco/occupational carcinogens, chronic inflammation and inherited pathways drive tumor growth and spread.

Determinants and modifiers. Tobacco, certain occupational exposures, age, family syndromes, chronic analgesic/inflammatory states and stochastic factors.

Assessment. Imaging, cystoscopy/urine evaluation, tissue and staging; kidney function and nephron-sparing considerations.

Intervention logic. Prevention/cessation, surgery or endoscopic therapy, radiation/systemic targeted/immune therapy and surveillance; preserve function when oncologically safe.

Known limits and open questions. Incidental small renal masses may be indolent; urinary biomarkers and screening lack universal utility.

Suite links. 04 cancer; 09 occupation; 14 oncology; 03F reserve.

Evidence anchors: ^[18,25,26]

Building a kidney and fluid-monitoring platform

Renal status changes over hours to years. The platform should distinguish filtration, injury, volume, urine output, electrolytes and treatment delivery rather than combining them into one kidney score.

6.1 Low-risk longitudinal stack

- Validated home blood pressure with technique and cuff metadata.
- Scale with stable placement and a defined edema/weight protocol.
- Medication, fluid, illness and symptom timeline.
- Laboratory import with specimen time, assay, eGFR equation and reference.
- Urine albumin/protein and urinalysis under standardized collection.
- Optional bladder ultrasound only with validated volume models and trained use.
- Dialysis telemetry separated from patient-reported recovery, symptoms and function.

Selected evidence anchors: [6,28]

6.2 Signals that must not be conflated

Signal	Represents	Does not prove
Creatinine rise	Change in production, distribution and/or clearance.	Direct structural injury, cause or exact GFR in nonsteady state.
Oliguria	Low urine output under current intake, hemodynamics and tract state.	Low GFR alone; obstruction and retention must be considered.
Albuminuria	Glomerular leak and/or handling under collection conditions.	A single disease or irreversible progression.
Weight gain	Net mass change, often fluid but also intake/body changes.	Intravascular volume or need for diuresis without exam/context.
Dialysis clearance number	Removal of a selected small-solute proxy.	Adequate volume, nutrition, function or patient experience.

Selected evidence anchors: [6,41]

Dialysis device safety architecture

Dialysis exposes blood or peritoneum to a complex machine and fluid system. Independent safety channels, fault detection, water quality and human factors are essential.

7.1 Critical hazards

- Air embolism, blood leak, line disconnection and access needle displacement.
- Incorrect conductivity, temperature, ultrafiltration or anticoagulation.
- Water microbial/chemical contamination and endotoxin.
- Hemolysis, clotting, pressure sensor or pump failure.
- Cybersecurity or remote-setting compromise.
- Alarm fatigue, bypass, confusing interface and inadequate home support.
- Loss of power, supply or emergency transport during home therapy.

Selected evidence anchors: [29,30,32,34]

Acid-base as a coupled-system problem

Acid-base interpretation should be a sequence: verify values and clinical state, identify primary processes, test expected compensation, correct for relevant proteins, and search for mixed causes across lungs, kidneys, GI tract, metabolism and toxins.

8.1 Minimum reasoning sequence

- Confirm pH and whether the blood gas/specimen is appropriate.
- Classify respiratory and metabolic directions.
- Calculate anion gap with albumin/context and compare changes where useful.
- Assess expected compensation without assuming it is complete.
- Use lactate, ketones, renal function, medication/toxin and GI history selectively.
- Treat the cause and dangerous physiology; do not chase a formula in isolation.

Selected evidence anchors: [3,7,35]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current renal and urinary health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

9.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current renal and urinary health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[41]

9.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[17]

9.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[3,5]

9.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[41]

9.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[17]

9.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

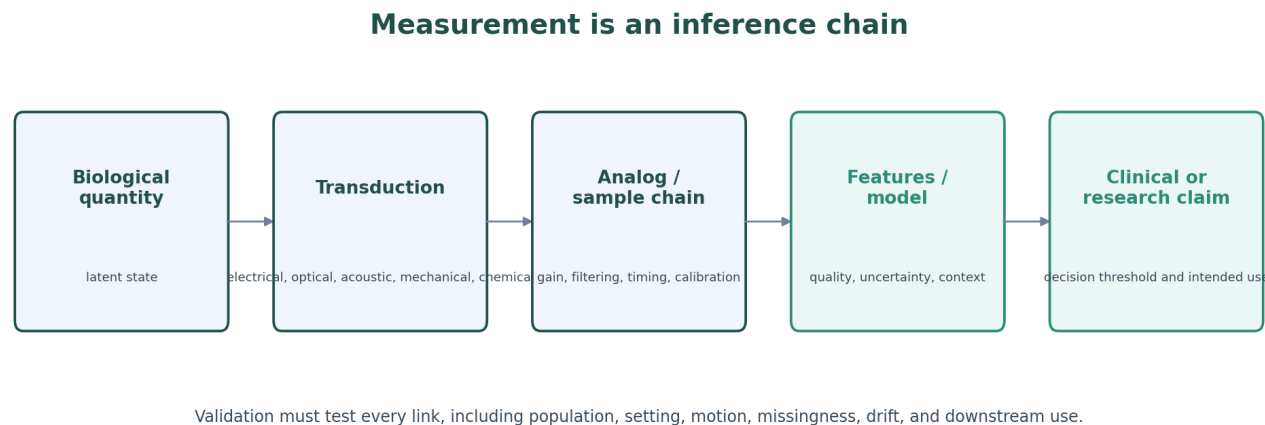


Figure 10.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
History, pressure, volume and fluid balance	Noninvasive	Symptoms, medication/exposure, pressure, weight, edema, orthostasis, intake/output and voiding pattern.	Urgency, volume state, chronicity, functional and causal hypotheses.	Immediate and longitudinally meaningful.	Intake/output and “dry weight” are imprecise; edema has multiple causes; pressure technique matters.
Serum creatinine and eGFR	Minimally invasive specimen	Enzymatic/Jaffe creatinine mapped through demographic equation to estimated GFR.	Filtration estimate and trajectory.	Widely available and outcome-linked.	Muscle mass, diet, secretion, drugs, assay and non-steady state; equation uncertainty at individual level.
Cystatin C and combined eGFR	Minimally invasive specimen	Protein filtered and metabolized by kidney, mapped to GFR equations.	Alternative/confirmatory filtration estimate.	Less dependent on muscle and combined equations improve average precision.	Affected by inflammation, thyroid, steroids, adiposity and assay; still an estimate.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Measured GFR and clearance tests	Minimally invasive to invasive	Timed plasma/urine disappearance or clearance of exogenous filtration markers.	Higher-fidelity GFR for selected decisions.	Better reference than routine estimates.	Time, sampling, compartment models, assay and urine-collection errors; burdensome.
Urinalysis dipstick	Noninvasive specimen	Colorimetric pads detect protein, blood, leukocyte esterase, nitrite, glucose, ketones, pH and concentration proxies.	Rapid screening pattern.	Low cost and point-of-care.	Concentration, timing, oxidants/reductants, pigments and limited sensitivity/specificity; “blood” is heme not necessarily RBCs.
Urine microscopy	Noninvasive specimen	Optical or automated imaging of cells, casts, crystals and organisms.	Localization clues for glomerular, tubular, infectious and crystal disease.	Direct morphology and high diagnostic value in context.	Collection delay, operator skill, variable definitions and automated misclassification.
Albumin/protein quantification	Noninvasive specimen	Immunologic or chemical assay in spot ratio or timed collection.	Albuminuria/proteinuria severity and trend.	Strong prognostic and treatment marker.	Exercise, fever, infection, menstruation, concentration and biologic variability; nonalbumin proteins need other tests.
Urine electrolytes, osmolality and metabolic collections	Noninvasive specimen	Ion-selective, osmometric and chemical assays in spot or timed urine.	Concentrating function, excretion, stone risk and selected acid-base/volume clues.	Direct output measurement when collection is valid.	Diuretics, nonsteady state and collection error; fractional indices are not universal diagnoses.
Renal and bladder ultrasound	Noninvasive acoustic	Anatomic imaging and bladder volume estimate.	Size, cysts, hydronephrosis, stones/masses, residual urine and procedure guidance.	Portable, no radiation and first-line for many questions.	Operator/body habitus, limited ureter/stone sensitivity, volume-model error and hydronephrosis-obstruction mismatch.
CT and CT urography	Noninvasive/minimally invasive ionizing	Unifactorially, cross-sectional X-ray with protocols for stones, masses, vessels or urothelium.	Stone/obstruction, trauma, mass, anatomy and staging.	High spatial resolution and rapid acute diagnosis.	Radiation, contrast, incidental findings and protocol dependence.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
MRI and MR urography	Noninvasive/minimally invasive	Magnetic anatomy, flow/perfusion/diffusion and urinary tract imaging.	Mass, vascular, cystic and selected functional/anatomic information.	No ionizing radiation and high soft-tissue contrast.	Time, motion, implants, contrast considerations and quantitative harmonization.
Nuclear renography	Minimally invasive ionizing	Radiotracer uptake and excretion over time.	Split kidney function, drainage and selected perfusion/obstruction assessment.	Functional information by kidney.	Hydration, bladder, timing, renal impairment and model assumptions; radiation.
Kidney biopsy	Invasive	Core tissue for light, immunofluorescence and electron/molecular pathology.	Glomerular/interstitial diagnosis and chronicity.	Mechanistic classification that can change therapy.	Bleeding, sampling error, observer/classification variability and static snapshot.
Bladder scan, uroflow and urodynamics	Noninvasive to minimally invasive	Ultrasound residual, urine flow, pressure catheters and pelvic EMG measure storage/voiding.	Residual, flow pattern, detrusor pressure/compliance and outlet coordination.	Mechanistic assessment for selected complex or invasive-treatment decisions.	Artificial setting, catheter effects, infection, poor symptom correlation and protocol dependence.
Continuous urine-output sensors	Minimally invasive catheter-associated or external	Optical, weight, flow, capacitance or imaging systems estimate urine output.	High-frequency output and trend.	Potential earlier oliguria detection and reduced manual burden.	Catheter infection/obstruction, bag position, motion, condensation, stool contamination and output not equal GFR.
Dialysis machine and water-system sensing	Invasive extracorporeal/peritoneal therapeutic	Pressure, flow, conductivity, temperature, air, blood leak, ultrafiltration and water quality sensors.	Treatment delivery, alarms, dose proxies and safety.	Real-time control of high-risk therapy.	Sensor drift, alarm fatigue, bypass, access recirculation, water contamination, cybersecurity and telemetry without context.
Novel injury and molecular biomarkers	Minimally invasive specimen	Proteins, metabolites, RNA or imaging features associated with tubular/glomerular injury or prognosis.	Earlier injury, mechanism or risk hypotheses.	May detect change before creatinine.	Context specificity, unclear actions, assay/platform differences and no proven utility for many markers.

10.0.1 History, pressure, volume and fluid balance

Measured quantity and transducer. Symptoms, medication/exposure, pressure, weight, edema, orthostasis, intake/output and voiding pattern.

Typical outputs. Urgency, volume state, chronicity, functional and causal hypotheses.

Strengths. Immediate and longitudinally meaningful.

Limitations and artifacts. Intake/output and “dry weight” are imprecise; edema has multiple causes; pressure technique matters.

Reference and validation strategy. Standardized measurement, repeated trends and corroboration with clinical/lab data.

Clinical/research maturity. Mature.

Evidence anchors: [2,6]

10.0.2 Serum creatinine and eGFR

Measured quantity and transducer. Enzymatic/Jaffe creatinine mapped through demographic equation to estimated GFR.

Typical outputs. Filtration estimate and trajectory.

Strengths. Widely available and outcome-linked.

Limitations and artifacts. Muscle mass, diet, secretion, drugs, assay and non-steady state; equation uncertainty at individual level.

Reference and validation strategy. Traceable assay, appropriate equation, trend and confirmation when decisions are sensitive.

Clinical/research maturity. Mature with known limits.

Evidence anchors: [6,31]

10.0.3 Cystatin C and combined eGFR

Measured quantity and transducer. Protein filtered and metabolized by kidney, mapped to GFR equations.

Typical outputs. Alternative/confirmatory filtration estimate.

Strengths. Less dependent on muscle and combined equations improve average precision.

Limitations and artifacts. Affected by inflammation, thyroid, steroids, adiposity and assay; still an estimate.

Reference and validation strategy. Assay standardization, intended-population equation and measured-GFR comparison.

Clinical/research maturity. Mature adjunct.

Evidence anchors: [6,10]

10.0.4 Measured GFR and clearance tests

Measured quantity and transducer. Timed plasma/urine disappearance or clearance of exogenous filtration markers.

Typical outputs. Higher-fidelity GFR for selected decisions.

Strengths. Better reference than routine estimates.

Limitations and artifacts. Time, sampling, compartment models, assay and urine-collection errors; burdensome.

Reference and validation strategy. Protocol and assay traceability, repeated sampling and uncertainty reporting.

Clinical/research maturity. Specialized reference method.

Evidence anchors: [6,31]

10.0.5 Urinalysis dipstick

Measured quantity and transducer. Colorimetric pads detect protein, blood, leukocyte esterase, nitrite, glucose, ketones, pH and concentration proxies.

Typical outputs. Rapid screening pattern.

Strengths. Low cost and point-of-care.

Limitations and artifacts. Concentration, timing, oxidants/reductants, pigments and limited sensitivity/specificity; “blood” is heme not necessarily RBCs.

Reference and validation strategy. Controls, expiration/storage, timing and microscopy/quantification confirmation.

Clinical/research maturity. Mature screening.

Evidence anchors: [8,31]

10.0.6 Urine microscopy

Measured quantity and transducer. Optical or automated imaging of cells, casts, crystals and organisms.

Typical outputs. Localization clues for glomerular, tubular, infectious and crystal disease.

Strengths. Direct morphology and high diagnostic value in context.

Limitations and artifacts. Collection delay, operator skill, variable definitions and automated misclassification.

Reference and validation strategy. Fresh standardized specimen, training, interobserver and clinical/biopsy correlation.

Clinical/research maturity. Mature specialist/basic method.

Evidence anchors: [5,6]

10.0.7 Albumin/protein quantification

Measured quantity and transducer. Immunologic or chemical assay in spot ratio or timed collection.

Typical outputs. Albuminuria/proteinuria severity and trend.

Strengths. Strong prognostic and treatment marker.

Limitations and artifacts. Exercise, fever, infection, menstruation, concentration and biologic variability; nonalbumin proteins need other tests.

Reference and validation strategy. Repeat confirmation, assay QC and context; timed collection selectively.

Clinical/research maturity. Mature.

Evidence anchors: [6,9]

10.0.8 Urine electrolytes, osmolality and metabolic collections

Measured quantity and transducer. Ion-selective, osmometric and chemical assays in spot or timed urine.

Typical outputs. Concentrating function, excretion, stone risk and selected acid-base/volume clues.

Strengths. Direct output measurement when collection is valid.

Limitations and artifacts. Diuretics, nonsteady state and collection error; fractional indices are not universal diagnoses.

Reference and validation strategy. Collection completeness, timing, medication context and mass-balance checks.

Clinical/research maturity. Mature with interpretation limits.

Evidence anchors: [3,8]

10.0.9 Renal and bladder ultrasound

Measured quantity and transducer. Anatomic imaging and bladder volume estimate.

Typical outputs. Size, cysts, hydronephrosis, stones/masses, residual urine and procedure guidance.

Strengths. Portable, no radiation and first-line for many questions.

Limitations and artifacts. Operator/body habitus, limited ureter/stone sensitivity, volume-model error and hydronephrosis-obstruction mismatch.

Reference and validation strategy. Phantom/volume validation, standardized views and cross-imaging/reference.

Clinical/research maturity. Mature.

Evidence anchors: [8,30]

10.0.10 CT and CT urography

Measured quantity and transducer. Cross-sectional X-ray with protocols for stones, masses, vessels or urothelium.

Typical outputs. Stone/obstruction, trauma, mass, anatomy and staging.

Strengths. High spatial resolution and rapid acute diagnosis.

Limitations and artifacts. Radiation, contrast, incidental findings and protocol dependence.

Reference and validation strategy. Dose/phantom QA, renal-risk assessment and clinical/pathology reference.

Clinical/research maturity. Mature.

Evidence anchors: [25,34]

10.0.11 MRI and MR urography

Measured quantity and transducer. Magnetic anatomy, flow/perfusion/diffusion and urinary tract imaging.

Typical outputs. Mass, vascular, cystic and selected functional/anatomic information.

Strengths. No ionizing radiation and high soft-tissue contrast.

Limitations and artifacts. Time, motion, implants, contrast considerations and quantitative harmonization.

Reference and validation strategy. Phantom/protocol QA, cross-reference and outcome validation.

Clinical/research maturity. Mature specialized.

Evidence anchors: [25,40]

10.0.12 Nuclear renography

Measured quantity and transducer. Radiotracer uptake and excretion over time.

Typical outputs. Split kidney function, drainage and selected perfusion/obstruction assessment.

Strengths. Functional information by kidney.

Limitations and artifacts. Hydration, bladder, timing, renal impairment and model assumptions; radiation.

Reference and validation strategy. Tracer/camera QC, standardized protocol and surgical/clinical reference.

Clinical/research maturity. Mature specialist.

Evidence anchors: [8,31]

10.0.13 Kidney biopsy

Measured quantity and transducer. Core tissue for light, immunofluorescence and electron/molecular pathology.

Typical outputs. Glomerular/interstitial/vascular diagnosis and chronicity.

Strengths. Mechanistic classification that can change therapy.

Limitations and artifacts. Bleeding, sampling error, observer/classification variability and static snapshot.

Reference and validation strategy. Image guidance, specimen adequacy, standardized pathology and clinical/genetic integration.

Clinical/research maturity. Mature reference with limits.

Evidence anchors: [5,6]

10.0.14 Bladder scan, uroflow and urodynamics

Measured quantity and transducer. Ultrasound residual, urine flow, pressure catheters and pelvic EMG measure storage/voiding.

Typical outputs. Residual, flow pattern, detrusor pressure/compliance and outlet coordination.

Strengths. Mechanistic assessment for selected complex or invasive-treatment decisions.

Limitations and artifacts. Artificial setting, catheter effects, infection, poor symptom correlation and protocol dependence.

Reference and validation strategy. Volume/pressure/flow calibration, standardized maneuvers and outcome prediction.

Clinical/research maturity. Mature specialized.

Evidence anchors: [8,16]

10.0.15 Continuous urine-output sensors

Measured quantity and transducer. Optical, weight, flow, capacitance or imaging systems estimate urine output.

Typical outputs. High-frequency output and trend.

Strengths. Potential earlier oliguria detection and reduced manual burden.

Limitations and artifacts. Catheter infection/obstruction, bag position, motion, condensation, stool contamination and output not equal GFR.

Reference and validation strategy. Known-flow bench tests, catheter fault injection, clinical comparison and utility trial.

Clinical/research maturity. Emerging adjunct.

Evidence anchors: [28,34]

10.0.16 Dialysis machine and water-system sensing

Measured quantity and transducer. Pressure, flow, conductivity, temperature, air, blood leak, ultrafiltration and water quality sensors.

Typical outputs. Treatment delivery, alarms, dose proxies and safety.

Strengths. Real-time control of high-risk therapy.

Limitations and artifacts. Sensor drift, alarm fatigue, bypass, access recirculation, water contamination, cybersecurity and telemetry without context.

Reference and validation strategy. Traceable calibration, fault injection, independent safety channels, human factors and postmarket surveillance.

Clinical/research maturity. Mature regulated system.

Evidence anchors: [29,30,34]

10.0.17 Novel injury and molecular biomarkers

Measured quantity and transducer. Proteins, metabolites, RNA or imaging features associated with tubular/glomerular injury or prognosis.

Typical outputs. Earlier injury, mechanism or risk hypotheses.

Strengths. May detect change before creatinine.

Limitations and artifacts. Context specificity, unclear actions, assay/platform differences and no proven utility for many markers.

Reference and validation strategy. Analytical validation, external prognostic validation and randomized biomarker-guided utility trials.

Clinical/research maturity. Research to selected adjunct.

Evidence anchors: [40,41,43]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise renal and urinary health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Continued on next page

Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

11.0.1 Intended use and measurand

Design requirements. Define the precise renal and urinary health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [28,32]

11.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [30,32]

11.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

11.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[31,33]

11.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[31]

11.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[40]

11.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[40]

11.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[19,29,30]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Blood pressure, diabetes and vascular prevention	Primary/secondary prevention	Reduce glomerular and vascular injury through pressure, glucose, tobacco, lipids and activity management.	Strong evidence for slowing CKD and reducing cardiovascular events with appropriately selected treatments.	Hypotension, electrolyte/AKI risk, polypharmacy and inequitable access.	Home/clinic pressure quality, eGFR/albuminuria, glucose, potassium, symptoms and events.
RAAS blockade and antiproteinuric therapy	Pharmacology	Reduce intraglomerular pressure, proteinuria and maladaptive hormonal signaling.	Improves kidney and cardiovascular outcomes in defined albuminuric/heart-failure populations.	Creatinine/potassium changes, hypotension, pregnancy risk and unsafe combinations.	Pressure, creatinine/eGFR and potassium after initiation/change, albuminuria and symptoms.
SGLT2 and cardiorenal-metabolic therapy	Pharmacology	Modify proximal transport, tubuloglomerular feedback and systemic metabolism to protect kidney/heart in eligible populations.	Reduces CKD progression and heart-failure events across several diabetes and nondiabetes groups.	Genital infection, volume effects, ketoacidosis risk and peri-illness/surgery management.	eGFR trajectory, volume/pressure, adverse effects, glucose/ketones when indicated and outcomes.
Volume, electrolyte and acid-base management	Supportive/pharmacology	Adjust osmotic, diuretics, binders, alkali, replacement and offending drugs to restore safe physiology.	Essential for symptoms and prevention of arrhythmia, neurologic injury and pulmonary edema.	Overcorrection, dehydration, drug interactions, GI burden and treating numbers without cause.	Weight/pressure, labs at appropriate intervals, symptoms, ECG and underlying disease.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Immunologic and cause-specific kidney therapy	Pharmacology/biology	Biologically defined immune, infectious, genetic, metabolic or obstructive causes.	Can preserve kidney when diagnosis and timing are correct; benefit varies by pathology and risk.	Infection, malignancy, organ toxicity, fertility and cost; incorrect empiric immune therapy can harm.	Disease activity, kidney function/proteinuria, drug/toxicity, infection and pathology-specific markers.
Medication stewardship and nephrotoxin prevention	System/pharmacology	Ecologic prescriptions/OTC/supplements, adjust for kidney function and avoid harmful combinations or dehydration contexts.	Prevents common AKI, accumulation and adverse drug events.	Undertreatment from overly conservative dosing and fragmented prescribers.	Medication list, kidney trajectory, drug levels selectively, adverse events and transitions of care.
Stone prevention and urologic procedure	Behavior/pharmacology/procedure	Increase/avoid volume safely, modify specific solute risks and remove/fragment obstructing stones when needed.	Reduces recurrence when matched to stone/urine phenotype; procedures relieve obstruction.	Diet burden, electrolyte effects, infection, bleeding, ureter injury and incomplete clearance.	Stone events/analysis, 24-hour urine, imaging, kidney function and adherence.
Urinary infection prevention and treatment	Etiologic/systemic	Use indicated antibiotics and remove catheter; obstruction or reflux contributors.	Effective when true infection is present; stewardship prevents resistance and unnecessary harm.	Resistance, allergy/toxicity, C. difficile, recurrent overdiagnosis and catheter harm.	Symptoms, culture/susceptibility, recurrence, catheter days and adverse events.
Pelvic-floor and bladder rehabilitation	Rehabilitation/behavior	Bladder training, pelvic-floor coordination, mobility/toileting adaptation and self-catheter education when needed.	Improves selected incontinence/voiding phenotypes and participation.	Incorrect strengthening can worsen dyssynergia; burden, infection and access.	Bladder diary, residual, leakage, function, goals and quality of life.
Hemodialysis and peritoneal dialysis	Organ support	Remove solute/fluid and correct electrolytes/acid-base through extracorporeal membrane or peritoneum.	Life-sustaining for kidney failure; modality should align with physiology, life and goals.	Hypotension, infection, access, peritonitis, protein loss, treatment burden and device/water failures.	Symptoms/function, volume, labs, access/peritoneum, machine data, residual function and patient goals.
Kidney transplantation	Organ replacement	Implant a donor kidney and maintain immune tolerance pharmacologically.	Generally offers longer survival and quality for eligible kidney-failure patients compared with chronic dialysis.	Surgery, rejection, infection, malignancy, drug toxicity, donor risk and allocation inequity.	Graft function/protein, drug exposure, rejection/infection, cardiovascular/metabolic effects and function.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Conservative kidney management and palliative care	Whole-person care	Treat symptoms and support goals without dialysis or alongside limited interventions when burden exceeds benefit.	Appropriate for selected people and should be an active care pathway, not abandonment.	Late discussion, prognostic uncertainty, family conflict and unequal service access.	Symptoms, function, goals, caregiver burden, crisis planning and place of care.

12.0.1 Blood pressure, diabetes and vascular prevention

Target and mechanism. Reduce glomerular and vascular injury through pressure, glucose, tobacco, lipids and activity management.

Evidence and expected outcome. Strong evidence for slowing CKD and reducing cardiovascular events with appropriately selected treatments.

Risks, burden, and equity. Hypotension, electrolyte/AKI risk, polypharmacy and inequitable access.

Response monitoring. Home/clinic pressure quality, eGFR/albuminuria, glucose, potassium, symptoms and events.

Evidence anchors: [2,6,9]

12.0.2 RAAS blockade and antiproteinuric therapy

Target and mechanism. Reduce intraglomerular pressure, proteinuria and maladaptive hormonal signaling.

Evidence and expected outcome. Improves kidney and cardiovascular outcomes in defined albuminuric/heart-failure populations.

Risks, burden, and equity. Creatinine/potassium changes, hypotension, pregnancy risk and unsafe combinations.

Response monitoring. Pressure, creatinine/eGFR and potassium after initiation/change, albuminuria and symptoms.

Evidence anchors: [2,6]

12.0.3 SGLT2 and cardiorenal-metabolic therapy

Target and mechanism. Modify proximal transport, tubuloglomerular feedback and systemic metabolism to protect kidney/heart in eligible populations.

Evidence and expected outcome. Reduces CKD progression and heart-failure events across several diabetes and nondiabetes groups.

Risks, burden, and equity. Genital infection, volume effects, ketoacidosis risk and peri-illness/surgery management.

Response monitoring. eGFR trajectory, volume/pressure, adverse effects, glucose/ketones when indicated and outcomes.

Evidence anchors: [2,6,9]

12.0.4 Volume, electrolyte and acid-base management

Target and mechanism. Adjust intake, diuretics, binders, alkali, replacement and offending drugs to restore safe physiology.

Evidence and expected outcome. Essential for symptoms and prevention of arrhythmia, neurologic injury and pulmonary edema.

Risks, burden, and equity. Overcorrection, dehydration, drug interactions, GI burden and treating numbers without cause.

Response monitoring. Weight/pressure, labs at appropriate intervals, symptoms, ECG and underlying disease.

Evidence anchors: [3,6,34]

12.0.5 Immunologic and cause-specific kidney therapy

Target and mechanism. Treat defined immune, infectious, genetic, metabolic or obstructive causes.

Evidence and expected outcome. Can preserve kidney when diagnosis and timing are correct; benefit varies by pathology and risk.

Risks, burden, and equity. Infection, malignancy, organ toxicity, fertility and cost; incorrect empiric immune therapy can harm.

Response monitoring. Disease activity, kidney function/proteinuria, drug/toxicity, infection and pathology-specific markers.

Evidence anchors: [6,15,23]

12.0.6 Medication stewardship and nephrotoxin prevention

Target and mechanism. Reconcile prescriptions/OTC/supplements, adjust for kidney function and avoid harmful combinations or dehydration contexts.

Evidence and expected outcome. Prevents common AKI, accumulation and adverse drug events.

Risks, burden, and equity. Undertreatment from overly conservative dosing and fragmented prescribers.

Response monitoring. Medication list, kidney trajectory, drug levels selectively, adverse events and transitions of care.

Evidence anchors: [14,34,39]

12.0.7 Stone prevention and urologic procedure

Target and mechanism. Increase urine volume safely, modify specific solute risks and remove/fragment obstructing stones when needed.

Evidence and expected outcome. Reduces recurrence when matched to stone/urine phenotype; procedures relieve obstruction.

Risks, burden, and equity. Diet burden, electrolyte effects, infection, bleeding, ureter injury and incomplete clearance.

Response monitoring. Stone events/analysis, 24-hour urine, imaging, kidney function and adherence.

Evidence anchors: [8,34]

12.0.8 Urinary infection prevention and treatment

Target and mechanism. Use indicated antibiotics and remove catheter, obstruction or reflux contributors.

Evidence and expected outcome. Effective when true infection is present; stewardship prevents resistance and unnecessary harm.

Risks, burden, and equity. Resistance, allergy/toxicity, C. difficile, recurrent overdiagnosis and catheter harm.

Response monitoring. Symptoms, culture/susceptibility, recurrence, catheter days and adverse events.

Evidence anchors: [20,21]

12.0.9 Pelvic-floor and bladder rehabilitation

Target and mechanism. Bladder training, pelvic-floor coordination, mobility/toileting adaptation and self-catheter education when needed.

Evidence and expected outcome. Improves selected incontinence/voiding phenotypes and participation.

Risks, burden, and equity. Incorrect strengthening can worsen dyssynergia; burden, infection and access.

Response monitoring. Bladder diary, residual, leakage, function, goals and quality of life.

Evidence anchors: [16,17]

12.0.10 Hemodialysis and peritoneal dialysis

Target and mechanism. Remove solute/fluid and correct electrolytes/acid-base through extracorporeal membrane or peritoneum.

Evidence and expected outcome. Life-sustaining for kidney failure; modality should align with physiology, life and goals.

Risks, burden, and equity. Hypotension, infection, access, peritonitis, protein loss, treatment burden and device/water failures.

Response monitoring. Symptoms/function, volume, labs, access/peritoneum, machine data, residual function and patient goals.

Evidence anchors: [6,30]

12.0.11 Kidney transplantation

Target and mechanism. Implant a donor kidney and maintain immune tolerance pharmacologically.

Evidence and expected outcome. Generally offers longer survival and quality for eligible kidney-failure patients compared with chronic dialysis.

Risks, burden, and equity. Surgery, rejection, infection, malignancy, drug toxicity, donor risk and allocation inequity.

Response monitoring. Graft function/protein, drug exposure, rejection/infection, cardiovascular/metabolic effects and function.

Evidence anchors: [6,15,38]

12.0.12 Conservative kidney management and palliative care

Target and mechanism. Treat symptoms and support goals without dialysis or alongside limited interventions when burden exceeds benefit.

Evidence and expected outcome. Appropriate for selected people and should be an active care pathway, not abandonment.

Risks, burden, and equity. Late discussion, prognostic uncertainty, family conflict and unequal service access.

Response monitoring. Symptoms, function, goals, caregiver burden, crisis planning and place of care.

Evidence anchors: [17,36,37]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can AKI biomarkers improve outcomes?	Several markers detect tubular stress or injury before or beyond creatinine in selected settings.	Whether biomarker-guided actions improve patient-important outcomes across routine care.	No single injury mechanism or intervention follows from a positive result; reference and thresholds vary.	Randomized biomarker-triggered care pathways with explicit actions, harms and cost.
What is the best measure of volume status?	History, examination, weight, pressure, ultrasound, biomarkers and device signals each provide partial information.	A reliable cross-setting individual reference for intravascular and tissue volume that guides therapy.	Fluid compartments, venous pressure, permeability and adaptation differ by heart/liver/kidney state.	Multimodal prospective intervention studies with hemodynamic and outcome anchors.
When should dialysis start and how intensive should it be?	Life-threatening refractory physiology and uremic symptoms are accepted indications; one eGFR threshold is insufficient.	Optimal timing and prescription for heterogeneous frailty, residual function and goals.	Selection, competing risk, treatment burden and device-centered metrics complicate trials.	Pragmatic preference-sensitive trials and registries with function, symptoms, caregiver burden and survival.
Can home urine and electrolyte sensors be trustworthy?	Electrochemical and optical sensing is feasible for selected analytes.	Long-term calibration, contamination resistance and clinical utility outside controlled collection.	Variable pH, ionic strength, temperature, biofilm, cleaning and incomplete void capture.	Bench challenge matrices followed by reference-comparison and utility studies.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within renal and urinary health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Continued on next page

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

13.0.1 Can AKI biomarkers improve outcomes?

Established knowledge. Several markers detect tubular stress or injury before or beyond creatinine in selected settings.

Unresolved issue. Whether biomarker-guided actions improve patient-important outcomes across routine care.

Why current evidence is insufficient. No single injury mechanism or intervention follows from a positive result; reference and thresholds vary.

Research that would change confidence. Randomized biomarker-triggered care pathways with explicit actions, harms and cost.

Evidence anchors: [7,41]

13.0.2 What is the best measure of volume status?

Established knowledge. History, examination, weight, pressure, ultrasound, biomarkers and device signals each provide partial information.

Unresolved issue. A reliable cross-setting individual reference for intravascular and tissue volume that guides therapy.

Why current evidence is insufficient. Fluid compartments, venous pressure, permeability and adaptation differ by heart/liver/kidney state.

Research that would change confidence. Multimodal prospective intervention studies with hemodynamic and outcome anchors.

Evidence anchors: [2,6,40]

13.0.3 When should dialysis start and how intensive should it be?

Established knowledge. Life-threatening refractory physiology and uremic symptoms are accepted indications; one eGFR threshold is insufficient.

Unresolved issue. Optimal timing and prescription for heterogeneous frailty, residual function and goals.

Why current evidence is insufficient. Selection, competing risk, treatment burden and device-centered metrics complicate trials.

Research that would change confidence. Pragmatic preference-sensitive trials and registries with function, symptoms, caregiver burden and survival.

Evidence anchors: [6,36,37]

13.0.4 Can home urine and electrolyte sensors be trustworthy?

Established knowledge. Electrochemical and optical sensing is feasible for selected analytes.

Unresolved issue. Long-term calibration, contamination resistance and clinical utility outside controlled collection.

Why current evidence is insufficient. Variable pH, ionic strength, temperature, biofilm, cleaning and incomplete void capture.

Research that would change confidence. Bench challenge matrices followed by reference-comparison and utility studies.

Evidence anchors: [28,31]

13.0.5 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within renal and urinary health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[18]

13.0.6 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[40]

13.0.7 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[25,41]

13.0.8 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[28]

13.0.9 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[17]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Nonsteady filtration model	Can repeated creatinine, cystatin, urine output and context estimate changing GFR better than static equations?	Prospective acute-care cohort with measured filtration subsets, fluid balance and drug exposures; locked dynamical model.	Calibrated individual uncertainty and improved dosing/decision utility.	Model performance depends on retrospective documentation or fails during rapid change.
0. Fluid-status utility trial	Does a structured multimodal volume protocol improve outcomes over usual assessment?	Randomized pathway using standardized exam, weight, pressure, ultrasound and biomarkers with explicit treatment rules.	Reduced congestion/dehydration events and treatment burden without subgroup harm.	More testing with no net outcome benefit.
0. Home dialysis resilience study	Can home systems remain safe during power, network, supply and caregiver disruptions?	Simulation/fault injection plus prospective registry of device versions, support response and near misses.	Validated fallback designs and reduced serious incidents.	Safety depends on exceptional users or undocumented workarounds.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for renal and urinary health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

Continued on next page

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

14.0.1 0. Nonsteady filtration model

Question. Can repeated creatinine, cystatin, urine output and context estimate changing GFR better than static equations?

Design. Prospective acute-care cohort with measured filtration subsets, fluid balance and drug exposures; locked dynamical model.

Success criterion. Calibrated individual uncertainty and improved dosing/decision utility.

Failure or stop criterion. Model performance depends on retrospective documentation or fails during rapid change.

Evidence anchors: [7,40]

14.0.2 0. Fluid-status utility trial

Question. Does a structured multimodal volume protocol improve outcomes over usual assessment?

Design. Randomized pathway using standardized exam, weight, pressure, ultrasound and biomarkers with explicit treatment rules.

Success criterion. Reduced congestion/dehydration events and treatment burden without subgroup harm.

Failure or stop criterion. More testing with no net outcome benefit.

Evidence anchors: [2,42]

14.0.3 0. Home dialysis resilience study

Question. Can home systems remain safe during power, network, supply and caregiver disruptions?

Design. Simulation/fault injection plus prospective registry of device versions, support response and near misses.

Success criterion. Validated fallback designs and reduced serious incidents.

Failure or stop criterion. Safety depends on exceptional users or undocumented workarounds.

Evidence anchors: [29,32,34]

14.0.4 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for renal and urinary health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: [38]

14.0.5 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

14.0.6 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: [32,40]

14.0.7 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [41,42]

14.0.8 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: [17]

14.0.9 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: [38,42]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Pressure, perfusion, congestion, cardiorenal therapies and cardiovascular events.	volume, eGFR or treatment is interpreted in hemodynamic context.
03B	Respiratory Health	Acid-base compensation, pulmonary edema and ventilation in kidney failure.	dyspnea or blood gas changes occur.
03D	Endocrine and Metabolic Health	Diabetes, mineral, water, adrenal and potassium regulation.	metabolic disease causes CKD or changes renal physiology.
03E	Digestive and Liver Health	GI losses, malabsorption, cirrhosis/ascites and hepatorenal physiology.	fluid, stones or kidney failure is linked to digestive disease.
03G	Hematologic Health	Renal anemia, thrombosis, bleeding and dialysis access.	blood count/coagulation or erythropoietin is involved.
03C	Nervous System Health	Uremic encephalopathy, autonomic bladder and osmotic neurologic injury.	cognition, seizures or voiding reflects neural control.
10	Function, Disability and Rehabilitation	Dialysis burden, continence, mobility and conservative-care outcomes.	a laboratory endpoint must be translated into daily life.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Albuminuria	Urinary albumin excretion, usually expressed as a ratio or timed rate; it is variable and must be interpreted in context.
Estimated GFR	Population-equation estimate of filtration from endogenous markers; each individual value has uncertainty.
Hyperfiltration	Higher filtration at whole-kidney or single-nephron level that may be adaptive or maladaptive depending on context.
Oliguria	Low urine output over a stated interval; it is a sign, not a cause or direct GFR measurement.
Residual kidney function	Native renal clearance and urine output remaining during kidney-failure treatment, often clinically important.
Uremia	Clinical syndrome caused by advanced kidney dysfunction and interacting retained solutes/homeostatic failure, not one measured toxin.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

- [1] Technical Evidence and Systems Review. Atlas of Human Health: A Unified Taxonomy. 2026. Master taxonomy and conceptual architecture for this suite.
- [2] Technical Evidence and Engineering Review. Cardiovascular Health and Measurement Engineering. 2026. Volume 03A of this suite.
- [3] Boron WF, Boulpaep EL. Medical Physiology, 4th ed.. 2022.
- [4] Hall JE, Hall ME. Guyton and Hall Textbook of Medical Physiology, 15th ed.. 2025.
- [5] Kumar V, Abbas AK, Aster JC. Robbins and Cotran Pathologic Basis of Disease, 11th ed.. 2025.
- [6] Kidney Disease: Improving Global Outcomes. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of CKD. 2024. <https://kdigo.org/guidelines/ckd-evaluation-and-management/>
- [7] Kidney Disease: Improving Global Outcomes. KDIGO Clinical Practice Guideline for Acute Kidney Injury. 2012/updates. <https://kdigo.org/guidelines/acute-kidney-injury/>
- [8] National Institute of Diabetes and Digestive and Kidney Diseases. Kidney disease information and tests. 2026. <https://www.niddk.nih.gov/health-information/kidney-disease>
- [9] American Diabetes Association. Standards of Care in Diabetes - 2026. 2026. https://diabetesjournals.org/care/issue/49/Supplement_1
- [10] Endocrine Society. Clinical practice guidelines. 2026. <https://www.endocrine.org/clinical-practice-guidelines>
- [11] International Osteoporosis Foundation. Clinical resources and fracture risk assessment. 2026. <https://www.osteoporosis.foundation/health-professionals>
- [12] World Health Organization. Blood safety and availability. 2025. <https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability>
- [13] International Union of Basic and Clinical Pharmacology. IUPHAR/BPS Guide to Pharmacology. 2026. <https://www.guidetopharmacology.org/>
- [14] U.S. Food and Drug Administration, Office of Clinical Pharmacology. 2025 Annual Report. 2025. <https://www.fda.gov/media/191717/download>
- [15] World Health Organization. Roadmap for access to medicines, vaccines and health products 2025-2030. 2025. <https://www.who.int/teams/health-product-policy-and-standards/medicines-selection-ip>
- [16] World Health Organization. Rehabilitation 2030 initiative. 2026. <https://www.who.int/initiatives/rehabilitation-2030>
- [17] World Health Organization. International Classification of Functioning, Disability and Health. 2001/2026. <https://www.who.int/standards/classifications/international-classification-of-functioning>
- [18] World Health Organization. Precision medicine: a path towards targeted, personalized and equitable care (WHA79.8). 2026. <https://www.who.int/news/item/22-05-2026-world-health-assembly-endorses>

- [19] National Commission for the Protection of Human Subjects. The Belmont Report. 1979. <https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/>
- [20] World Health Organization. Global Action Plan on Antimicrobial Resistance 2026-2036. 2026. <https://www.who.int/news/item/25-05-2026-the-world-health-assembly-adopts-updated-global-action-plan-2026-2036>
- [21] Centers for Disease Control and Prevention. Antimicrobial resistance threats, surveillance, and prevention. 2025/2026. <https://www.cdc.gov/antimicrobial-resistance/>
- [22] National Institute of Allergy and Infectious Diseases. Research priorities in infection, immunity, and microbiome. 2025/2026. <https://www.niaid.nih.gov/research>
- [23] American College of Rheumatology. Clinical practice guidelines. 2026. <https://rheumatology.org/clinical-practice-guidelines>
- [24] American Society of Hematology. ASH clinical practice guidelines. 2026. <https://www.hematology.org/education/clinicians/guidelines-and-quality-care/clinical-practice-guidelines>
- [25] National Cancer Institute. Cancer biology, screening, diagnosis and treatment resources. 2026. <https://www.cancer.gov/>
- [26] Hanahan D. Hallmarks of Cancer: New Dimensions. 2022. <https://doi.org/10.1158/2159-8290.CD-21-1059>
- [27] Lopez-Otin C, Blasco MA, Partridge L, Serrano M, Kroemer G. Hallmarks of aging: an expanding universe. 2023. <https://doi.org/10.1016/j.cell.2022.11.001>
- [28] U.S. Food and Drug Administration. Digital Health Center of Excellence and current guidance. 2026. <https://www.fda.gov/medical-devices/digital-health-center-excellence>
- [29] U.S. Food and Drug Administration. Cybersecurity in Medical Devices: Quality Management System Considerations and Content of Premarket Submissions. 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-considerations-and-content-of-premarket-submissions>
- [30] International Electrotechnical Commission. IEC 60601-1: Medical electrical equipment - basic safety and essential performance. 2005+A1+A2. <https://webstore.iec.ch/publication/67497>
- [31] International Organization for Standardization. ISO 13485: Medical devices - quality management systems. 2016. <https://www.iso.org/standard/59752.html>
- [32] International Organization for Standardization. ISO 14971: Medical devices - application of risk management. 2019. <https://www.iso.org/standard/72704.html>
- [33] International Electrotechnical Commission. IEC 62304: Medical device software life-cycle processes. 2006+A1. <https://webstore.iec.ch/publication/6792>
- [34] World Health Organization. Global Patient Safety Action Plan 2021-2030. 2021. <https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan>
- [35] Singer M, Deutschman CS, Seymour CW, et al.. The Third International Consensus Definitions for Sepsis and Septic Shock. 2016. <https://doi.org/10.1001/jama.2016.0287>
- [36] World Health Organization. Palliative care. 2025. <https://www.who.int/news-room/fact-sheets/detail/palliative-care>
- [37] World Health Organization. WHOQOL: Measuring Quality of Life. 2026. <https://www.who.int/tools/whoqol>
- [38] World Health Organization. Health equity. 2026. <https://www.who.int/health-topics/health-equity>

- [39] Institute of Medicine. Best Care at Lower Cost: The Path to Continuously Learning Health Care in America. 2013. <https://doi.org/10.17226/13444>
- [40] Collins GS, Moons KGM, Dhiman P, et al.. TRIPOD+AI statement. 2024. <https://www.tripod-statement.org/>
- [41] Guyatt GH, Oxman AD, Vist GE, et al.. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. 2008. <https://doi.org/10.1136/bmj.39489.470347.AD>
- [42] Damschroder LJ, Reardon CM, Widerquist MAO, Lowery J. The updated Consolidated Framework for Implementation Research. 2022. <https://doi.org/10.1186/s13012-022-01245-0>
- [43] Chen C, Wang J, Pan D, et al.. Applications of multi-omics analysis in human diseases. 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10390758/>

Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.